

# PRABAL KANTI KHAN

## Embedded Systems Engineer

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### PROFESSIONAL SUMMARY

Embedded Systems Engineer (B.Tech ECE, 2026) who takes products end-to-end — from circuit design and bare-metal STM32/ESP32 firmware to power electronics, wireless communication and cloud dashboards. Strong in register-level Embedded C, I<sup>2</sup>C/SPI/UART, LoRa and IoT integration, backed by hands-on hardware bring-up, soldering and instrument-based debugging. Seeking an Embedded / Firmware Engineer role; open to relocation.

### TECHNICAL SKILLS

**Languages:** Embedded C, C, C++, Python, SQL, JavaScript

**MCUs & Hardware:** STM32 (F446RE, bare-metal), ESP32, Arduino, Raspberry Pi; schematic & PCB reading, soldering, SMPS, Li-ion, solar

**Protocols & Interfaces:** I<sup>2</sup>C, SPI, UART, GPIO, LoRa (RYLR998), MQTT, Wi-Fi, Bluetooth

**Firmware Concepts:** register-level programming (RCC, GPIO, MODER, ODR, IDR), interrupts & NVIC, peripheral drivers, ADC, PWM, data acquisition

**Tools & Diagnostics:** multimeter, oscilloscope, Git & GitHub, Arduino IDE, VS Code, Django, Linux

### EXPERIENCE

#### IoT & Django Intern — Euphoria GenX (via Heritage Institute of Technology) | Jun – Jul 2025

- Built real-time IoT dashboards linking live sensor data to a Django backend; validated end-to-end signal flow from sensor to visualisation.
- Traced and resolved faults across the data-acquisition and automation chain.

#### STEM Innovation Engineer — STEMROBO Technologies Pvt. Ltd. | Feb 2022 – Jul 2023

- Assembled, wired and functionally tested embedded hardware across 20+ innovation labs — MCU boards, relay panels, sensor/actuator modules.
- Performed component-level fault isolation on non-functioning equipment using multimeter and diagnostic instruments.
- Standardised build/test documentation; trained 500+ users with 90% workshop satisfaction.

### PROJECTS

#### LoRa Agriculture Automation — Team Leader · Final Year Project | 2025 – 2026

- Dual-node system: ESP32 + RYLR998 LoRa farm node reading DHT22 / MQ135 / soil-moisture and driving Peltier AC, humidifier, fan and irrigation pump via relays; portable OLED control node with Auto/Manual override.
- Designed the full power architecture (12V/20A SMPS, dual 18650 + solar/TP4056) for 24/7 off-grid operation; added an MQTT cloud dashboard.

#### Smart Home Automation (Arduino + Bluetooth) — Team Leader | 2022

- Multi-module relay panel — Bluetooth switching, IR doorbell, LDR lighting, automatic tank cutoff (BC547 + relay), solar powered; complete firmware written in Embedded C.

#### Additional Builds — documented on YouTube @TechnicZ

- DIY Bluetooth speaker (PVC enclosure + 4S Li-ion); portable Raspberry Pi 4B+ touch computer; custom 15.4Ah 18650 pack with BMS and an off-grid solar inverter.

### EDUCATION & CERTIFICATION

**B.Tech, Electronics & Communication Engineering** — Heritage Institute of Technology, Kolkata | 2026 · DGPA 7.98

**Diploma, Electronics & Instrumentation Engineering** — Siliguri Govt. Polytechnic | 2022 · GPA 7.7

**Certification:** Microcontroller Embedded C Programming — Udemy (FastBitLab, Kiran Nayak) | Jun 2026

### ADDITIONAL

Languages: Bengali (Native), English & Hindi (Fluent) | Batch 2026 · No active backlogs | Builds documented on YouTube @TechnicZ